Math Kangaroo Practice - Grades 3-4

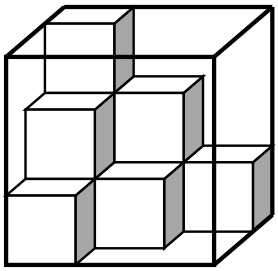
Practice set for 2007 #14

Each original problem is followed by three similar practice problems.

Problem 1: 2007 #14 - Cube Aquarium

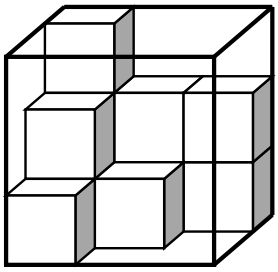
Arranging 1 dm cubes inside a 3 dm aquarium (counting 3D cubes / volume)

2007 #14 Daniella has an aquarium in the shape of a cube with edges 3 dm each. She started arranging cubes with edges 1 dm each inside the aquarium, in the way you can see on the picture. At most how many more such cubes can Daniella put into the aquarium?



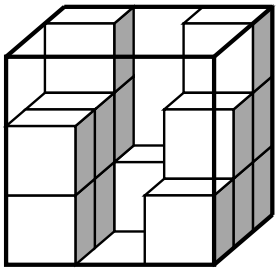
A) 9 **B)** 13 **C)** 17 **D)** 21 **E)** 27

Practice 1.1 Leo has an aquarium in the shape of a cube with edges 3 dm each. He started arranging cubes with edges 1 dm each inside the aquarium, in the way you can see on the picture. At most how many more such cubes can Leo put into the aquarium?



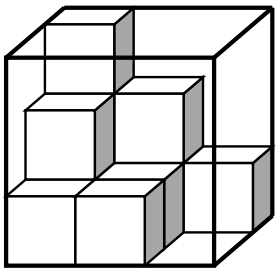
A) 16 **B)** 14 **C)** 12 **D)** 11 **E)** 9

Practice 1.2 Maya has an aquarium in the shape of a cube with edges 3 dm each. She started arranging cubes with edges 1 dm each inside the aquarium, in the way you can see on the picture. At most how many more such cubes can Maya put into the aquarium?



A) 17 **B)** 16 **C)** 15 **D)** 14 **E)** 13

Practice 1.3 Lucas has an aquarium in the shape of a cube with edges 3 dm each. He arranged some 1 dm cubes inside as shown in the picture. He wants to fill the aquarium completely. If the small cubes are sold in boxes of 3 cubes each, what is the least number of boxes Lucas needs to buy?



A) 5 **B)** 6 **C)** 7 **D)** 8 **E)** 9

Answer Key & Explanations

For practice_set_2.pdf (Practice set for 2007 #14)

Solutions & kid-friendly explanations

2007 #14: C The aquarium holds 3 × 3 × 3 = 27 small cubes in total. Counting the cubes already inside: the front layer has 1 cube, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 1 = 6 cubes, for a total of 1 + 3 + 6 = 10 cubes. Therefore, Daniella can put at most 27 - 10 = 17 more cubes into the aquarium.

1.1: A The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 1 cube, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 2 = 7 cubes, for a total of 1 + 3 + 7 = 11 cubes. Therefore, Leo can put at most 27 - 11 = 16 more cubes into the aquarium.

1.2: E The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 2 + 1 = 3 cubes, the middle layer has 2 + 2 = 4 cubes, and the back layer has 3 + 1 + 3 = 7 cubes, for a total of 3 + 4 + 7 = 14 cubes. Therefore, Maya can put at most 27 - 14 = 13 more cubes into the aquarium.

1.3: B The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 1 + 1 = 2 cubes, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 1 = 6 cubes, for a total of 2 + 3 + 6 = 11 cubes. Lucas needs 27 - 11 = 16 more cubes. Since each box contains 3 cubes, 5 boxes provide 15 cubes (which is not enough), so Lucas needs at least 6 boxes (providing 18 cubes).